

Q1(a)

$$\begin{aligned}\text{Elastic PE} &= \frac{1}{2} k x^2 \\ &= \frac{1}{2} (120)(10 \times 10^{-2})^2 \\ &= \mathbf{0.60 \text{ J}}\end{aligned}$$

Q1(b)

Using Law of Conservation of Energy, taking the lowest position of the box as corresponding to zero GPE,

$$\begin{aligned}\text{KE}_i + \text{GPE}_i + \text{EPE}_i &= \text{KE}_f + \text{GPE}_f + \text{EPE}_f \\ 0 + (2.0)(9.81)(D \sin 40^\circ) + 0 &= 0 + 0 \\ + \frac{1}{2} (120)D^2\end{aligned}$$

$$\mathbf{D = 0.210 \text{ m}}$$

[Alternative: loss of GPE of box = gain in elastic PE of spring]

Q1(c)

D will increase.

Q1(d)

Using Law of Conservation of Energy,

$$\begin{aligned}\text{KE}_i + \text{GPE}_i + \text{EPE}_i &= \text{KE}_f + \text{GPE}_f + \text{EPE}_f + W \text{ where } W = \text{work done to overcome friction} \\ 0 + (2.0)(9.81)(D' \sin 40^\circ) + 0 &= 0 + 0 \\ + \frac{1}{2} (120)D'^2 + (5.0)D'\end{aligned}$$

$$\mathbf{D' = 0.127 \text{ m}}$$

[Alternative: loss of GPE of box = gain in elastic PE of spring + work done to overcome friction]

Q1(e)

A spring with a larger spring constant will be stiffer.

This will cause D' to become smaller because the spring will stretch less.

Q2(a)